Introduction to HTML :-

→ Hypes text Mastup Language.
→ Website Dev.

Divisions :- Frontend - users can see on screen.
Backend - Program designing (codes), users can't see, they can just run or execute (Database set).

Client Side Lang. — Works on our devices ie. Bank account details.

Server Side Lang. — huge connection and our data is stored in some far away server base i.e. Eg. Admission form.
In ↓ PHP

→ Hypes Text :- A piece of text that works as a link.
→ Mastup Lang :- It is way of writing layout info within document.
→ It opens in Browsers, the browsers searches for HTML Code inside the text and use them to change layout, insert image or create links to other pages
→ HTML Docs are text files, so can be created or edited in a simple text editor.

→ WYSIWYG :- "What you see is what you get" editor which is editable like MS Word, instead of focusing on Code, it focuses on visuals.

→ Some Most popular editors :- Front Page or Dream Viewer or Notepad & Notepad++ etc.

→ HTML Acquires Forgiving Nature which means unlike any lang which gives you error if not written correctly it will just run the part of code which is correctly written and the tag/Attributes not closed or completed will not be shown on browser.

Basic Structure of HTML:-

```
<html>
    <head>                          → Info about file
        <Title>..</title>
    </head>
    <body>                          → Content of Doc.
        <h1-6>.....</h1-6>
    </body>
</html>
```

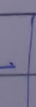
→ Tag are not Case-Sensitive

Characters formatting Tags:-

<B> ⇒ Bold ⇒ <B> Bold </B> = **Bold**
<U> ⇒ Underline ⇒ <U> Underline </U> = Underline [<ins>]
<I> ⇒ Italic ⇒ <I> Italic </I> = *Italics*

<Strike> ⇒ Strike ⇒ <Strike> Strike </strike> = ~~Strike~~ [<del>]

 ⇒ Line Break ⇒ 1
 2
 = 1
2
3

<hr> ⇒ Horizontal Line ⇒ 1 <hr> = 1


<sup> ⇒ Superscript ⇒ 3^{2} ⇒ 3 ² ⇒ 3²

<sub> ⇒ Subscript ⇒ (5) ₁₀ = (101) ₂

$= (5)_{10} = (101)_{2}$

Generation of HTML :-

1) HTML 1.0 :- (1993) Could link Document, Had fonts, Don't support tables.

2) HTML 2.0 :- (1996) Introduced "forms" and "tables"

3) HTML 3.2 :- (1997) Forms were more interactive.
Supports CSS [Cascading Style Sheet]
Has extended char. set.
Image handling got easier.

4) HTML 4.01 :- (1999) <header>, <footer> etc introduced.
Colspan, Rowspan used in tables.

5) XHTML 1.0 :- (2000) Extended Markup Lang. 1.0
Strict Standard.
Compatibility b/w Browsers.

6) HTML 5.0 :- (2014) Supports. Desktop & Mobiles
Email & Password Validation.
Audio & Video tags.
Section tag.
Latest version.

Shades of
#ff00ff - Pink
#ff0000 - Red
#00ff00 - Green
#0000ff - Blue
#ffff00 - Yellow

Font :-

Size

→ 1 → 7 (small → large)
3 (default)

Size is 1

Color

→ Name of Colour
or Hexadecimal Code
for Colour.

In Red Color

Face

→ Typeface of Font/Text
→ font should be installed [Both Dev. & Users]

In Arial font

Pre-tag : This tag keep the formatting as it is written in the HTML text editor.

<pre> Hey!
My name is Rehan.
At LMC in BCA
</pre>

⇒

Hey!
My name is Rehan.
At LMC in BCA.

Otherwise it would have looked like :-

Hey! My name is Rehan. At LMC in BCA.

<del> ⇒ Similar to <strike> tag.
<ins> ⇒ Similar to <U> tag.

Defining tags :-

<dl> :- start of the list
<dt> :- for Defining Term
<dd> :- for Definition of Term.
</dl> :- End of the list.

eg :- <html>
<body>
<dl> <dt> HTML </dt>
<dd> Hypes Text Markup Language </dd>
</body>

=> HTML
: Hypes Text Markup Language.

Abbreviation Tag :- "<abbr>"

```
<html>
  <body>
    <abbr title = "World Wide Web"> WWW </abbr>
    <br> <br>
    <acronym title = "World Wide Web"> WWW </acronym>
    <br> <br>
  </body>
</html>
```

#hr tag:- for Thematic break, used when topic is suddenly changed.
It draws a horizontal line b/w to topic or where you use this tag.

```
<body>
    <h2> Html </h2>
    <p> Hypest Text Mastup Language </p> <hr>

    <h2> hr tag </h2>
    <p> thematic horizontan line </p>
</body>
```

Output:-

HTML
Hypes Text Mastup Language

---

hr tag.
Thematic horizontal line.

Lists :- There are 3 types of lists:-

1.) Ordered List (<ol>) :- Displays items in numbered order (1,2,3...) (start from 1)
2) Unordered List (<ul>) :- Displays items in bulleted form (:--). (Generally)
3) Description List :- Displays Term & Def^{n}.

I <li> tag :- for List Items.

```
<body>
    <center> <h2> List of Courses </h2>
    </center>
<ol> <li> MCA </li>
    <li> BCA </li>
    <li> B.Sc. </li>
</ol>
</body>
```

⇒ Types of Ordered Lists :- We use type Attributes to specify numbering style.

- type = "1" ⇒ 1, 2, 3...
- type = "A" ⇒ A, B, C...
- type = "a" ⇒ a, b, c...
- type = "I" ⇒ I, II, III...
- type = "i" ⇒ i, ii, iii...

⇒ Start Attribute :- Specifies the starting of numbered list

- <ol start = "10"> ⇒ 10, 11, 12....

⇒ Value Attribute :- Changes the value of certain list item.

- <li value = "15"> English ⇒ 15. English.

```
<ol start = "10">
  <li> Hindi </li>
  <li value = "15"> English </li>
  <li value = "4"> Math </li>
</ol>
```

10. Hindi
⇒ 15. English
4. Math

III Types of unordered lists :-

- type = "Square" ⇒ ■
- type = "Disc" ⇒ • (Similar to bullet)
- type = "Circle" ⇒ ○

III Definition list or Description list :-

Tags :- `<dl>` Describe list — `<dt>` Describe term — `<dd>` Describe Data.

```
<body>
  <dl>
    <dt> BCA </dt>
    <dd> Bachelor of Computer Application </dd>
  </dl>
```

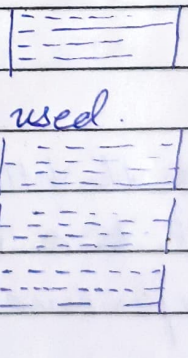
Nested List :-

```
<ul> <li> List Item 1 </li>
     <li> List Item 2 </li>
     <ul> <li> Item 3 </li>
          <li> Item 4 </li>
     </ul>
     <li> List </li>
     <ol> <li> List Item 5 </li>
          <li> List Item 6 </li>
     </ol>
</ul>
```

- List Item 1
- List Item 2
  - Item 3
  - Item 4
- List
  1. List Item 5
  2. List Item 6.

Text Alignment :- A formatting Attr. that determines alignment of text.

```
<p Align = "Left"> Left Alignment =>
This is default Alignment, even when not used.
<p Align = "Right"> Right Alignment =>
<p Align = "Center"> Center Alignment =>
<p Align = "Justify"> It stretches text towards both margin =>
```



#Head Section Detail:- It contains metadata (data about data).
→ Not directly displayed on the web pg.
→ Helps Browser and search engine under-standing web pg.

⇒ <style> Contains CSS rules for styling the HTML doc which is called internal CSS, useful for styles specific to a single page.

```
<html>
  <head>
    <style> p{color: red;}
    </style>
  </head>
  <body>
    <p> Hello </p>
  </body>
</html>
```

H.S.D defines the title of the Doc that appears in the browsers title bar; it is a mandatory and imp. element for SEO and user experience.

#Link Tag:- Create relationship b/w the current Doc & an external doc, it is most commonly used for external CSS (.CSS) file.

```
<head> <link rel="Styling Sheet"
        href="Styles.css">
</head>
```

Meta tag:- It provides invisible meta data that can't be defined by other tags.

Common Attributes :-

1) Char set :- Character encoding for the doc, Modern Std. = "UTF-8" used to ensure all char. are displayed correctly.

eg:- <meta charset = "UTF-8">

2) Name Attribute :- Specify name of m.d. used with the content att. to provide info like keyword or description

3) Content Attribute :- Contains value assigned with the name att. or HTTP-equiv.

eg:- <meta name = "Description" Content = "Overview of HTML">
<meta name = "Keyword" Content = "HTML, Head.">

4) Http-equiv :- <meta http-equiv = "refreshing" content = "30">
This will Auto-refresh the page every 30 sec.

<script> - This tag is used to embed a executable side script, most commonly Java script with an Html Doc, it serves as the primary mechanism for adding interactive & Dynamic behaviour to a Wb.pg.

```
<html>
    <head>
        <title> Script Program. </title>
    </head>
    <body> <script> alert ("Hello, World!"); </script>
    </body>
</html>
```

Nested element :- One element in other, creating a heirarchy. This str. is fundamental for organising content on Wb.pg. and define relationship b/w different element. The enclosing element is parent, while the element inside is child. child can also be a parent element.

```
<tag1>
    <tag2>
        <tag3>
        </tag3>
    </tag2>
</tag1>
```

Block-Level Element :- It always start on a new line, and the browser add some space before & after the element.

→ A Block level element always take the full width available (stretches out to the left & right as far as it can.

eg:- <p>,

Inline - Element :- an inline element does not start on a new line, an inline element only take up as much width as neccessary.

eg:- <img>, <small>, <dd>, <dt>, <pre>.

```
<html>
  <head>
      <title> Block-head - inline element </title>
  </head>
  <body>
      <p> Hello </p>
      <p> HTML </p>
      <img src = "img1.jpg">
      <img src = "img2.jpg">
  </body>
</html>
```

Container & Empty Tag :- Container tag requires both opening & closing tag. (→ eg:- <p>, <pre>, etc.)
Empty tag only have opening tag.
↳ eg:-
, <hr>, <img>.

Video Content (Video Tag) :- used to insert video file in HTML, supports multiple attributes.

<src> ⇒ Specifies the path to video file.

<control> ⇒ add vide controls like :- Play, Pause, Volume etc.

<autoplay> ⇒ auto plays the video when Web page reloads.

<loop> ⇒ repeat the video after it end.

<muted> ⇒ mute the video by default.

<poster> ⇒ Specifies an image to be displayed before the video starts.
Thumbnail

<width & height> ⇒ specify the dimensions of the video.

<source> ⇒ allows you to specify alternative video file, which the browser may choose from, the browser will use the first recognised format, the text :- "Browser don't support the video element" will be displayed.

```
<body>
  <video width="320" height="240" muted control loop
    poster="img.jpg"><source src="test.mp4"
    type="video/mp4">
    your browser doesn't support video tag.
  </video>
</body>
```

Audio Tag :- This tag is used to insert audio file in an html document. It also supports multiple attribute.

i) src ⇒ specifies the path of the audio file.
ii) Control ⇒ Add audio controls like, play, pause etc.
iii) autoplay ⇒ automatically plays audio when webpage loads.
iv) Loop ⇒ Repeats audio after it ends.
v) muted ⇒ mute the audio files. by default
vi) Pre-load ⇒ specify it and how the audio should be loaded when the page loads. preload attribute can help following values :- a) none :- This is the default value it indicates that browser should not preload the audio file; and it will start downloading them when user initiate playback.

b) metadata :- This value tells the browser to preload only the metadata of audio file, such as its duration and basic info. about the audio; this can be useful if you want to display audio duration to the user without fully loading audio data.

c) Auto ⇒ This value instructed the browser to preload the entire audio file as much as possible without delaying the loading of others important page content.
Audio tag contains one or more source tag with different audio sources. the text b/w audio and closing audio tag will be displayed in browsers that don't support the audio element.

```
<html>
  <body>
    <h1> The audio element </h1>
    <p> click on play button </p>
  <audio preload = "auto" controls>
  <source src = "audio.wav" type = "audio/wav">
    your browser doesn't support audio element.
  </audio>
  </body>
</html>
```

Anchor tag :- (<a> tag) :- This tag is used to create a hyper link on the webpage, this hyper link is used to link the webpage to other webpage or some section of same webpage. It is either used to provide an reference as its href value the syntax of <a> tag is :-

```
<a href = "Link"> Link name </a>
```

→ Title att. defines the title of the link, which appear to the user as a tool-tip.

→ Target attribute : is used to specify where to open the link document.

Syntax:-

<a target = "-blank/-self/-parent/-top/-frame name")

i) -Blank => opens the link document in a new tab or window.

ii) -Self => open the link in current frame.
iii) -parent => open the link in parent frame.
iv) - Top => open the link in top-most frame.
v) - framename => open the link in the named frame.

 Link name

```
<html>
  <body>
  <a href = "ABCD.html" title = "open Web page">
      click here
  </a>
</body>
</html>
```

→ Target attribute :- <a target = "-blank/-self/-parent/-top/-framename">

```
<html>
  <body> <a href = "ABCD.html" title = "A Webpage"
          target = "-blank"> <click here </a>
  </body>
</html>
```

→ Name tag :- This enables user to jump to specific section of a webpage. This is useful for navigating across large pages, the below program will jump from top to bottom, and vice-versa i.e. linking within the same page.

```
<body>
  <a name = "Top"> Top of Page </a> <br>
  <a href = "#Bottom"> Jump to Bottom </a> <br>
  <p> _ _ _
```

→ Images as Hyperlink :- We can lin 1 page to another by using an img as link, we use the img tag to display an image the <a> tag is used to create a Hyperlink, placing an <img> tag inside a tag display a clickable img. that leads to another page.

```
<body>
  <a href = "ABCD.html"> <img src = "img.jpg"> </a>
</body>
```

Tables :- It is an arrangement of data in rows and column. They is widely used in communication, research & data analysis.

<table> :- It is used to create table.
→ It is a container of table that specifies where the table begins & ends.

```
<table borde = "1" >  </table>
```

table row ⇒ <tr> ⇒ It defines a row in a table.
table heading ⇒ <th> ⇒ It defines table header, generally top row of the table.
table cell ⇒ <td> ⇒ It defines table data, table cell stores data to be displayed in the table.

```
<table border="1">
  <tr>
    <th></th>
    <th>column heading 1</th>
    <th>" " 2</th>
    <th>" " 3</th>
  </tr>
  <tr>
    <th>Row Heading 1</th>
    <td>Data</td>
    <td>D</td>
    <td>D</td>
  </tr>
  <tr>
    <th>Row Heading 2</th>
    <td>D</td>
    <td>D</td>
    <td>D</td>
  </tr>
  <tr>
    <th>Row Heading 3</th>
    <td>D</td>
    <td>D</td>
    <td>D</td>
  </tr>
</table>
```

	Column Heading 1	Column Heading 2	Column Heading 3
Row Header 1	Data	D	D
Row Header 2	D	D	D
Row Header 3	D	D	D